

Architectural Evaluation of Google Cloud Platform vs. Microsoft Azure: A Comparative Study of Real-Time Data Processing Performance

Cheryl Cheong Kah Voon
Faculty of Computing
University Technology of Malaysia
Johor, Malaysia
cherylcheongkah@graduate.utm.my

Abstract—In the era of big data, real-time data processing is a critical requirement for modern enterprises. This paper compares and analyzes Google Cloud Platform (GCP) and Microsoft Azure, focusing on their architectural suitability for high-speed data streaming and large-scale analytics. The study examines key technical dimensions, including global versus regional network models, serverless execution efficiency and integration with managed AI/ML services. By evaluating the trade-offs between GCP’s unified global infrastructure and Azure’s enterprise-grade integration framework, this research explores how different architectural philosophies address challenges related to latency, scalability and compliances. This analysis aims to provide a strategic framework for platform selection across various scenarios.

Index Terms—cloud computing, Google Cloud Platform, Microsoft Azure, real-time data processing, scalability

I. INTRODUCTION

The exponential growth of data from IoT, mobile and enterprise application has driven a surge in demand for real-time data processing. Modern organizations now require platforms capable of ingesting, processing and analyzing massive streams of data with minimal latency to enable timely decision-making, predictive analytics and automation [1]. Cloud platforms have emerged as the primary backbone for these workloads, offering managed services and elastic scalability that traditional on-premises system struggle to match. However, while cloud environment offer superior flexibility, their performance must be continuously evaluated against local cluster using standardized benchmarks to ensure they meet the compact demands of multi-node data processing [2].

Microsoft Azure and Google Cloud Platform (GCP) are the two leading providers in this space, each offering a specialized suite of service for real-time workloads. However, their architecture philosophies differ fundamentally. GCP was built to handle Google’s large-scale search and machine learning tools, while Azure is designed with enterprise users at the center, prioritizing seamless integration with existing Microsoft ecosystem. These differences in “architecture DNA” significantly impact application design, cost and operational efficiency, making the platform selection process a complex challenge for IT architects [3].

This paper aims to provide a comparative architectural and performance analysis of Azure and GCP, focusing on their suitability for real-time data processing workloads. This paper is organized as follows: Section II discusses real-time data processing concept, Section III explains the Google Cloud Platform, Section IV explains the Microsoft Azure, Section V provides a comparative analysis between two cloud services and Section VI concludes the paper.

II. OVERVIEW OF REAL-TIME DATA PROCESSING

Real-time data processing also known as stream processing refers to the continuous ingestion and analysis of data with near-zero latency. Unlike traditional batch processing that execute tasks in high-time intervals, real-time systems can process data “dynamically”. This is essential for uncovering hidden patterns in complex datasets before they lose their strategic value [4]. The efficiency of these system depends on the 4Vs of big data, where Velocity is the critical factor for enabling predictive analytics and automation [1].

A typical real-time pipeline follows a four-stage process which are ingestion (via brokers like Pub/Sub), transformation (cleaning and aggregation), Storage (high-speed analytics) and action (visualization or alerts). While these architecture are powerful, maintaining fault tolerance and managing high-speed ingestion across multi-node cluster remain challenging. The research by [2] emphasizes that as these system scale, rigorous performance benchmarking is essential to address the “complex realities” of modern enterprise workloads.

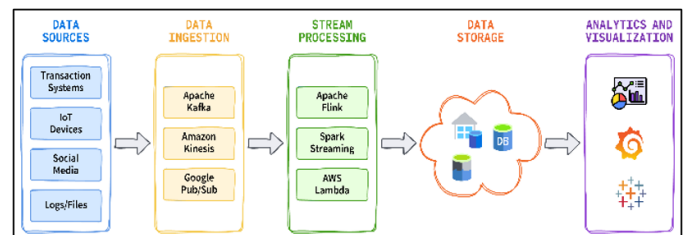


Fig. 1. Real-time Data Processing Process On Cloud Platform

III. GOOGLE CLOUD PLATFORM

Google Cloud Platform (GCP) is a suite of advanced cloud computing services that provides Infrastructure as a Service (IaaS), Platforms as a Service (PaaS) and serverless computing. It operates on a pay-as-you-go services such as virtual machines, storage, databases and advance tools for AI or Machines Learning, big data and networking [5]. GCP leveraging Google's global infrastructure, including 35 regions and high-speed fiberoptic network, to deliver scalable and reliable services [6].

A. Architecture

GCP's architecture is unique in that it uses a global software-defined network that treats the entire platform as a unified system rather than a series of isolated regions [7]. At the heart of this design is a global Virtual Private Cloud (VPC) that covers all geographic areas via a private submarine fiber optic network [6]. Unlike traditional regional models, this infrastructure ensures that data transmission bypasses the public internet, minimizing hop counts and latency and providing a high-speed backbone for distributed applications [7]. As shown in Fig. 2, the pipeline follows an event-driven process starting from Google Cloud Storage (GCS), which triggers a notification to Cloud Pub/Sub. Dataflow uses the Apache Beam model to extract and transform these messages, then inserts them into BigQuery for analysis, while managing raw data partitions and dead letter queues to ensure data integrity.

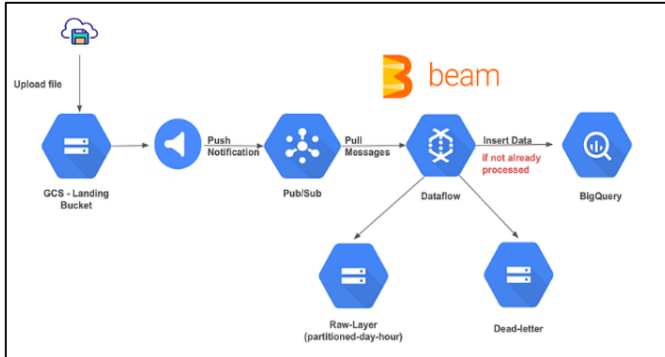


Fig. 2. GCP Real-time Data Pipeline on Google Cloud Platform

For real-time pipeline, GCP adopts a “serverless-first” approach, minimizing operational overhead while maintaining extremely high scalability [8]. This pipeline is centered around Cloud Pub/Sub, a global message broker capable of receiving millions of events per second with extremely high elasticity [6]. Data is then processed through Dataflow, a streaming and batch processing managed service based on the unified Apache Beam Model [7]. Finally, data is stored in BigQuery, a serverless data warehouse that separates computation from storage, achieving sub-second query performance even when handling petabyte-scale datasets [6].

B. Key Features

GCP provides several key features designed to enhance performance, scalability and intelligent data processing. One of its key advantages is high throughput and low latency as the infrastructure is optimized for intensive streaming tasks and rapid data manipulation. With its globally distributed architecture, dynamically allocates resources, handling everything from routine workloads to extreme traffic spikes without human intervention, ensuring exceptional scalability.

Furthermore, GCP improves operational efficiency by optimizing the cold start speed of serverless functions, crucial for the immediate responsiveness of event-driven systems. The platforms also provide seamless AI/ML integration, embedding tools like Vertex AI directly into data pipelines for real-time predictive modeling. Finally, GCP integrates robust, multi-layered security mechanisms at every level, providing default encryption and robust access management to ensure data security throughout its entire lifecycle.

C. Real-world study

Google leverages GCP's infrastructure to support YouTube, the world's largest video streaming platform serving over 2 billion monthly logged in users. Before integrating with GCP infrastructure, YouTube operated on independent internal system. Since 2021, Google has strategically migrated YouTube to the Google Cloud Platform to demonstrate GCP's ability to critical business workloads. YouTube now uses multiple GCP services, including Cloud Pub/Sub for event intake, Bigtable for metadata storage, Cloud Storage for video content and BigQuery for analysis. This architecture enables YouTube to handle billions of hours of video viewing every day while maintaining service reliability. Migration reflects how GCP's global network and serverless technology supports extreme scale and latency-sensitive workloads that exceed the requirements of traditional enterprises [9].

D. Advantage and Limitations

GCP offers significant advantages in scalability, performance and cost-effectiveness, leveraging the same underlying architecture as global services like YouTube and Google Map. Its pay-as-you-go model and serverless option like Cloud Functions reduce upfront capital expenditure and simplify deployment in data-intensive industries. Furthermore, GCP provides a superior ecosystem for data analytics and artificial intelligence through tools like BigQuery and Vertex AI, while maintaining a multi-layered security framework that comprehensively protects data security from the hardware layer to the application layer.

Besides these strength, GCP also has limitations, particularly regarding data privacy and potential vendor lock-in. Storing sensitive information in third-party environments can raise regulatory compliance issues and using proprietary APIs can make migrating to other platforms technically difficult. Furthermore, while the platform is cost-effective, managing fluctuating costs and addressing the complexities of multi-cloud interoperability requires specialized expertise. Certain

applications may also experience latency issues in specific high-precision domains, necessitating a strategic assessment of regional availability and technical support.

IV. MICROSOFT AZURE

Microsoft Azure is comprehensive cloud platform that provides Infrastructure as a Service (IaaS), Platform as a Service (PaaS) and Software as a Service (SaaS). It enables organizations to develop, deploy and manage a wide range of applications, from mobile and IoT solutions to high-performance computing, through a global network of data centres managed by Microsoft. Azure operates over 200 services in 140 countries and leverage interconnected infrastructure designed for data resilience to ensure high availability and disaster recovery capabilities [10] [11].

A. Architecture

Azure's architecture is geographically distributed across regions, availability zones and region pairs to ensure maximum uptime and data sovereignty [10]. Unlike a purely global network, this regional model allows for resource isolation, thereby meeting stringent regulatory requirements while maintaining high-speed connectivity within the infrastructure.

As shown in Fig. 3, the platform supports complex real-time and batch data flows. Streaming data sources such as fare and trip data are received through Azure Event Hub and processed instantly by Azure Stream Analytics. This "hot path" ensures low-latency analytics before data is store in Azure Cosmos DB, enabling sub-second retrieval. The final analysis results are visualized through Power BI, while Azure Monitor tracks system health to maintain its reliability.

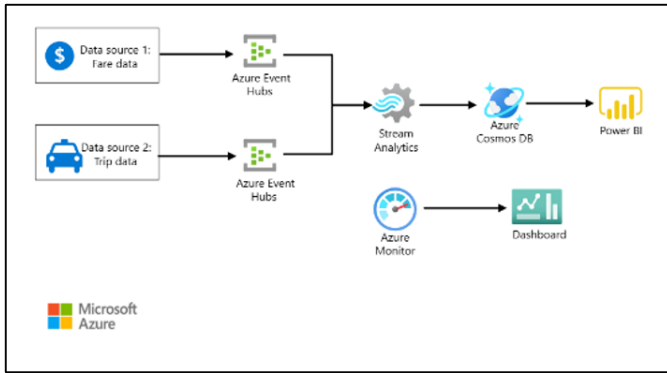


Fig. 3. Azure Real-time Streaming Architecture for Fleet Management

For large-scale data engineering, Azure integrates compute and storage through Azure Synapse Analytics and Azure Data Lake Storage. This architecture supports hyperscale distributed file service and multi-tiered applications, bridging the gap between big data exploration and traditional data warehouses [12].

B. Key Features

Microsoft Azure provides several key features designed to enhance performance, scalability and intelligent data processing capabilities. Its main advantages lie in high throughput

and low latency, enabling rapid data processing and efficient resource utilization even for high-performance computing task [13]. With its globally distributed architecture, Azure support dynamic horizontal scaling to handle varying workloads and extreme traffic spikes [11].

Furthermore, Azure improves operational efficiency by providing serverless data processing and optimized storage options such as Azure Blob Storage and SQL Database, meeting the needs of both unstructured and relational data [14]. The platforms also provides seamless AI/ML integration through Azure Machine Learning and Cognitive Services, embedding advanced analytics directly into the data lifecycle [15]. Finally, robust multi-layered security measures are integrated at each level, including identity protection and global compliance standards to ensure data security [16].

C. Real-world study

Walmart modernized its global retail operations through a strategic partnership with Microsoft, leveraging Azure Arc to manage over 300000 servers across its edge sites. Between 2023 and 2024, Walmart invested approximately \$580 million in Azure service to support large-scale data processing [17]. This deployment enabled centralized management and real-time inventory tracking, demonstrating Azure's ability to handle high-volume, distributed enterprise workloads while driving digital innovative in the retail industry [18].

D. Advantage and Limitations

Azure offers significant advantages in cost-effectiveness due to its global reach, extensive service offerings and pay-as-you-go pricing model [10] [19]. Its deep integration with existing Microsoft technologies makes it a top choice for enterprises seeking a familiar environment for cloud transformation [20]. Furthermore, the platform provides a unified environment for real-time IoT data and hyperscale big data storage, giving it a unique competitive advantage [11].

However, Azure also has some limitations. For example, its learning curve is relatively steep for developers unfamiliar with its vast ecosystem [21]. Some studies have indicated that Azure's dynamic horizontal scaling may be slower than that of its specialized competitors in certain distributed task scenarios [22]. Furthermore, managing and predicting the costs of fluctuating workloads requires expertise to ensure that the cloud environment remains cost-effective in the long term [19]; [10].

V. COMPARATIVE ANALYSIS

A. Comparison Table

In this section, comparison between GCP and Azure is performed.

B. Discussion

The comparison between GCP and Azure highlights their unique advantages in architecture and operations. GCP's global VPC architecture provides a unified, high-speed backbone network, minimizing latency and enabling superior

TABLE I
COMPARISON BETWEEN GCP AND AZURE

Dimension	GCP	Azure
Performance	Superior in latency-sensitive tasks and real-time analytics	Leading in ML inference time and throughput
Network Architecture	Global VPC	Regional VNet
Data Processing Model	Serverless-first (Pub/Sub & Dataflow)	Hybrid Integrated (Event Hubs & Synapse)
IoT & Edge	Effective for latency-sensitive hybrid models	Best for real-time IoT with high integration
Scalability	Exceptional in block and archive storage scaling	Strong integration with enterprise ecosystems

performance in high-speed real-time analytics. In contrast, Azure's regional VNet model prioritizes data sovereignty and geographic compliance, providing a more controlled environment for enterprises with stringent regulatory requirements.

Regarding processing models, GCP's "serverless-first" approach is optimized for rapid, automatic scaling during peak traffic periods. Azure's "hybrid integration" model, build a seamless bridge between on-premises systems and cloud-native services like Synapse. While GCP excels in specialized technology scaling, Azure remains a leader in enterprise ecosystem integration, providing a stable and easy-to-understand infrastructure for large-scale organizational deployments due to its superior data sovereignty, regulatory compliance features, and seamless integration with existing Microsoft enterprise ecosystems.

VI. CONCLUSION

In summary, GCP performs better in high-speed real-time analysis and global data synchronization due to its serverless-first nature and unifies global network. It is the ideal choice for latency-sensitive applications like autonomous driving and high-frequency trading. However, for enterprise environment with high governance requirements, Microsoft Azure remains the preferred choice, especially for industries like banking and healthcare. Azure's regional architecture and strict data residency controls are specifically designed for regulated industries. Banking and healthcare sectors require strictly compliance with data protection regulations, which Azure's infrastructure supports through its region-based data sovereignty model. Future research could explore multi-cloud strategies that combining GCP's analytics capabilities with Azure's enterprise-grade security to build a resilient and flexible data-driven infrastructure.

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